```
!pip install requests beautifulsoup4 pandas seaborn matplotlib
```

```
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (2.32.4)
Requirement already satisfied: beautifulsoup4 in /usr/local/lib/python3.12/dist-packages (4.13.5)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-packages (0.13.2)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests) (3.4.6)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests) (2026.2.25)
Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.12/dist-packages (from beautifulsoup4) (2.8.3)
Requirement already satisfied: typing-extensions>=4.0.0 in /usr/local/lib/python3.12/dist-packages (from beautifulsoup4) (4
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.62.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.5.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (26.0)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1
```

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
import requests
from bs4 import BeautifulSoup
import pandas as pd
import time

headers = {"User-Agent": "Mozilla/5.0"}

jobs = []

roles = ["data analyst", "python developer"]
location = "India"

for role in roles:
    url = f"https://www.linkedin.com/jobs-guest/jobs/api/seeMoreJobPostings/search?keywords={role}&location={location}&sta

    response = requests.get(url, headers=headers)
    soup = BeautifulSoup(response.text, "html.parser")

    job_cards = soup.find_all("div", class_="base-card")

    for job in job_cards:
        title = job.find("h3")
        company = job.find("h4")
        loc = job.find("span", class_="job-search-card__location")
        salary = job.find("span", class_="job-search-card__salary-info")

        jobs.append([
            title.text.strip() if title else None,
            company.text.strip() if company else None,
            loc.text.strip() if loc else None,
            salary.text.strip() if salary else "Not Disclosed",
            role
        ])

    time.sleep(2)

df = pd.DataFrame(jobs, columns=["Title", "Company", "Location", "Salary", "Role"])
df.head()
```

	Title	Company	Location	Salary	Role
0	Business Intelligence	MakeMyTrip	Gurugram, Haryana, India	Not Disclosed	data analyst
1	Data Analyst	Curefit	Bengaluru, Karnataka, India	Not Disclosed	data analyst
2	Data Analyst	Navi	Bangalore Urban, Karnataka, India	Not Disclosed	data analyst
3	Data Analyst	L&T Technology Services	Bengaluru, Karnataka, India	Not Disclosed	data analyst
4	Data Analyst	Persistent Systems	Pune District, Maharashtra, India	Not Disclosed	data analyst

Next steps: Generate code with df New interactive sheet

```
skills_list = [
    "python", "sql", "excel", "machine learning",
    "power bi", "tableau", "java", "c++", "aws"
]

def extract_skills(text):
    text = str(text).lower()
    return [skill for skill in skills_list if skill in text]

df["Skills"] = df["Title"].apply(extract_skills)
df.head()
```

	Title	Company	Location	Salary	Role	Skills
0	Business Intelligence	MakeMyTrip	Gurugram, Haryana, India	Not Disclosed	data analyst	[]
1	Data Analyst	Curefit	Bengaluru, Karnataka, India	Not Disclosed	data analyst	[]
2	Data Analyst	Navi	Bangalore Urban, Karnataka, India	Not Disclosed	data analyst	[]
3	Data Analyst	L&T Technology Services	Bengaluru, Karnataka, India	Not Disclosed	data analyst	[]
4	Data Analyst	Persistent Systems	Pune District, Maharashtra, India	Not Disclosed	data analyst	[]

Next steps: Generate code with df New interactive sheet

```
df_exploded = df.explode("Skills")
df_exploded = df_exploded.dropna()
df_exploded = df.explode("Skills").dropna()
df_exploded.head()
```

	Title	Company	Location	Salary	Role	Skills
10	Python developer	Infosys	Bengaluru East, Karnataka, India	Not Disclosed	python developer	python
11	Python Developer	Infosys	Bengaluru East, Karnataka, India	Not Disclosed	python developer	python
12	Python developer	Infosys	Hyderabad, Telangana, India	Not Disclosed	python developer	python
13	Python Developer	Infosys	Bengaluru, Karnataka, India	Not Disclosed	python developer	python
14	Python Developer	KPMG India	Bengaluru, Karnataka, India	Not Disclosed	python developer	python

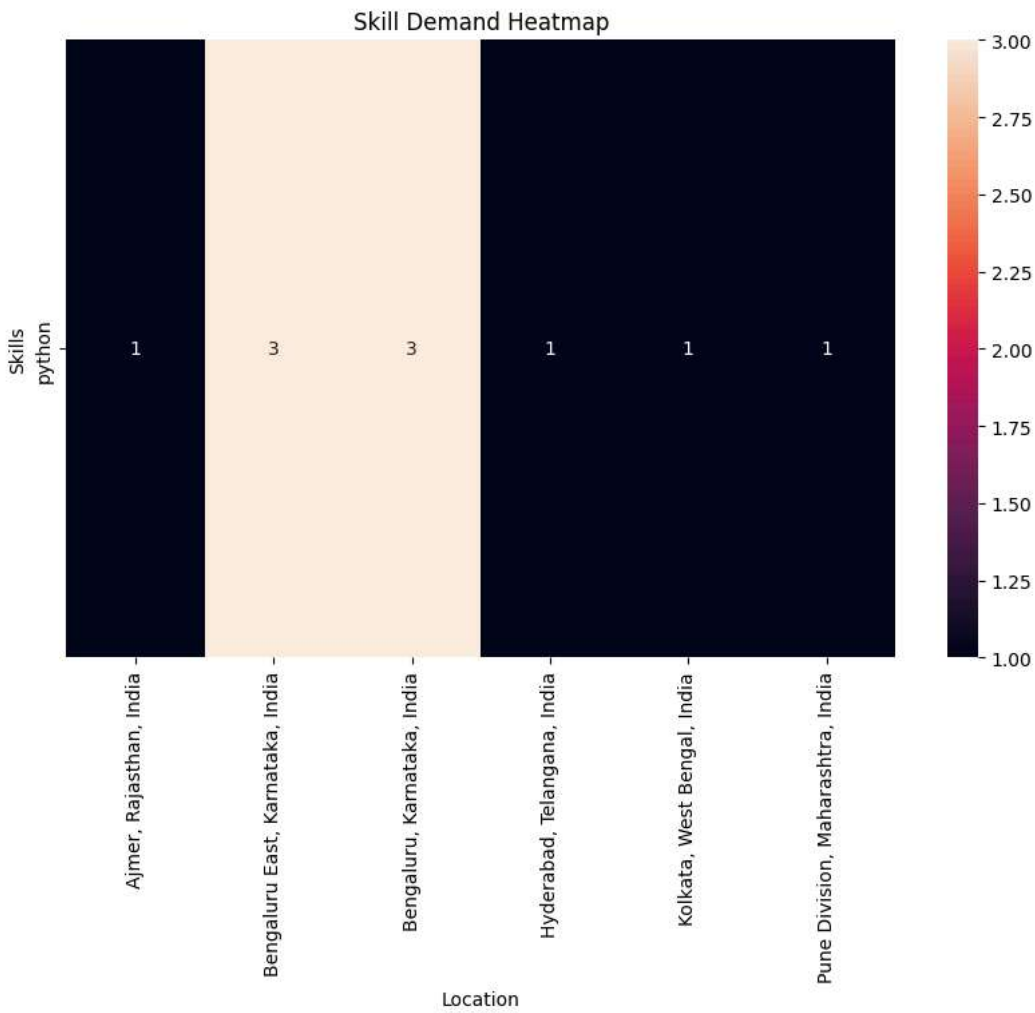
Next steps: Generate code with df_exploded New interactive sheet

```
skill_city = df_exploded.groupby(["Location", "Skills"]).size().reset_index(name="Count")

top_skills = skill_city.sort_values(["Location", "Count"], ascending=[True, False])
top_skills.head(10)
```

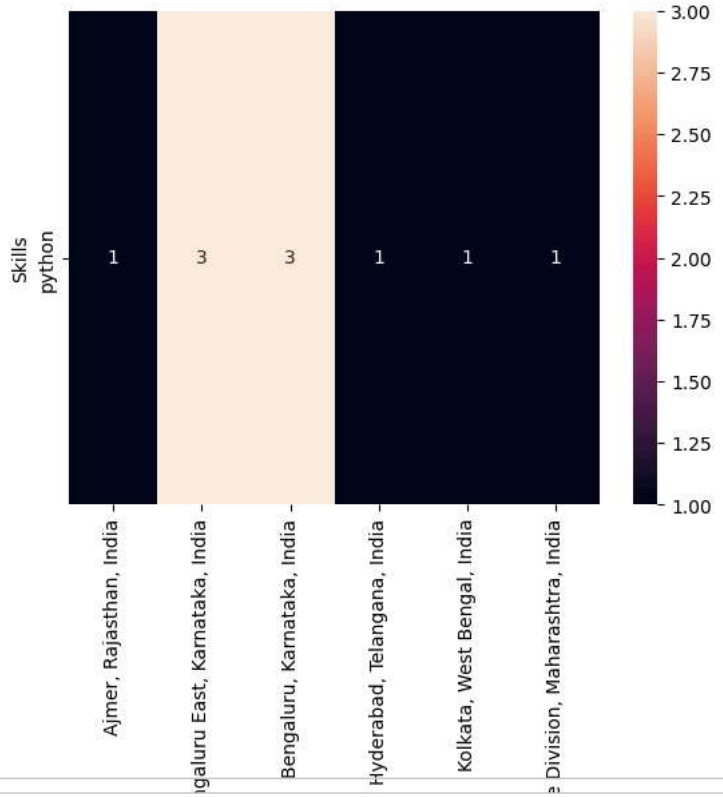
	Location	Skills	Count
0	Ajmer, Rajasthan, India	python	1
1	Bengaluru East, Karnataka, India	python	3
2	Bengaluru, Karnataka, India	python	3
3	Hyderabad, Telangana, India	python	1
4	Kolkata, West Bengal, India	python	1
5	Pune Division, Maharashtra, India	python	1

Next steps: Generate code with top_skills New interactive sheet

```
import seaborn as sns
import matplotlib.pyplot as plt

pivot = skill_city.pivot(index="Skills", columns="Location", values="Count").fillna(0)

plt.figure(figsize=(10,6))
sns.heatmap(pivot, annot=True)
plt.title("Skill Demand Heatmap")
plt.show()
```




```
pivot = df_exploded.pivot_table(index="Skills", columns="Location", aggfunc="size", fill_value=0)

sns.heatmap(pivot, annot=True)
plt.show()
```




```
pivot_role = df_exploded.pivot_table(index="Skills", columns="Role", aggfunc="size", fill_value=0)
pivot_role
```

Role	python developer
Skills	
python	10

```
df["Salary"].value_counts()
```

	count
Salary	
Not Disclosed	20

dtype: int64

```
df.to_csv("linkedin_jobs_data.csv", index=False)
df_exploded.to_csv("linkedin_jobs_skills.csv", index=False)
```